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Zener diode as voltage Regulation

Aim:

To simulate the line and load regulation operation zener diode

Apparatus Required

Laptop with Proteus software

Theory

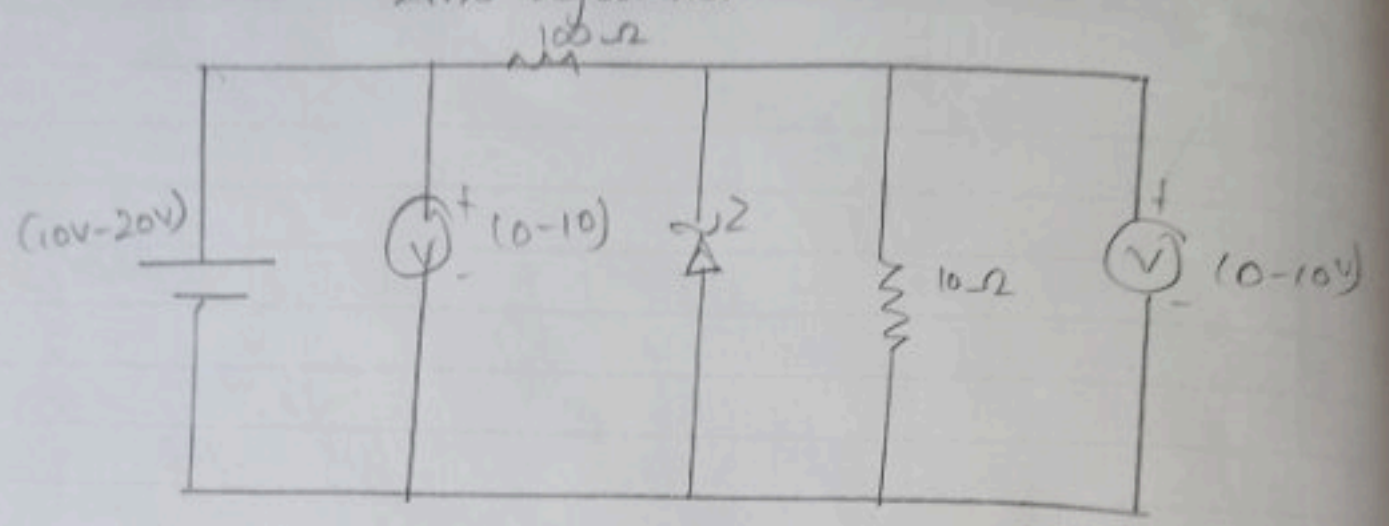
A heavily dropped PN Junction diode is called zener diode. Due to heavily dopped nature zener diode works under forward bias and reverse bias conditions

Zener diode under reverse bias condition is used as a voltage regulation. Even if the input voltage changes the output is constant. This is called line regulation

If the load resistance change the output voltage is connected this is called load regulation

Circuit diagram

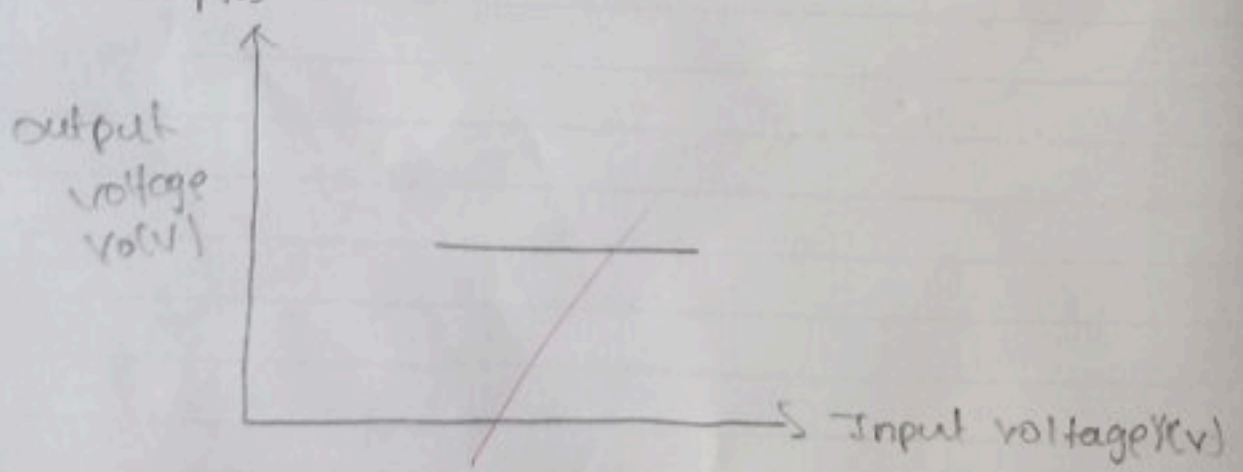
Line regulation



Tabulation

S.No	Input voltage V_i (V)	Output voltage V_o (V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6.	+17.0	+5.19

Model Graphs



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Procedure

1) Drag the required Component from the Proteus library.

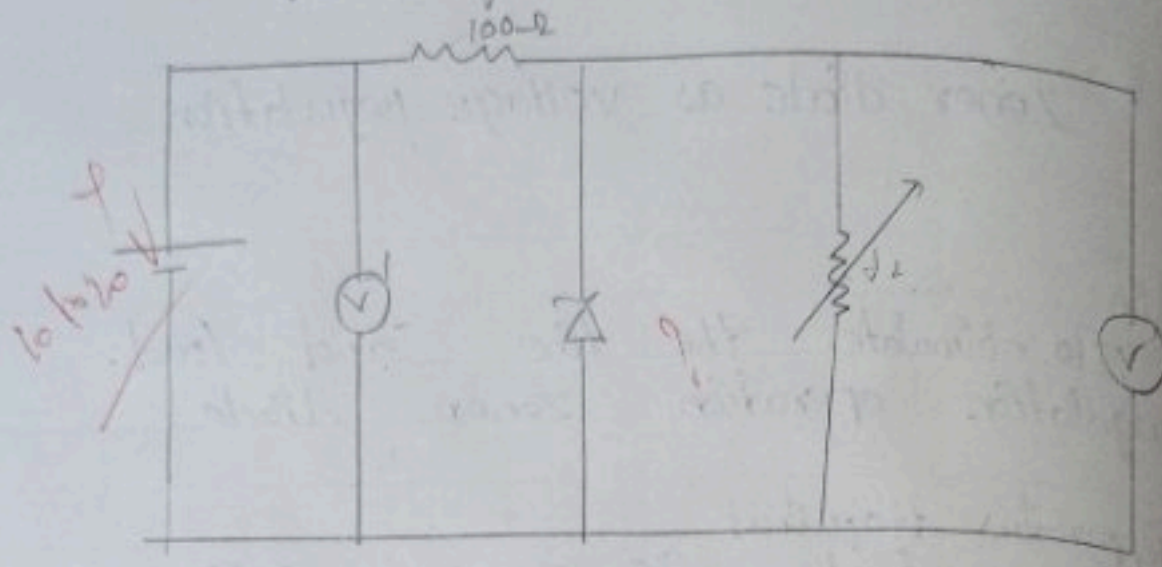
2) Connect the Component (required) as per circuit diagram

3) For line regulation vary the input voltage and not the outputed voltage in tabulation. Draw the graph between input voltage and output voltage

4) For load regulations vary the load resistance and the output voltage in tabulation Draw the graph between load resistance and output voltage.

Circuit Diagram

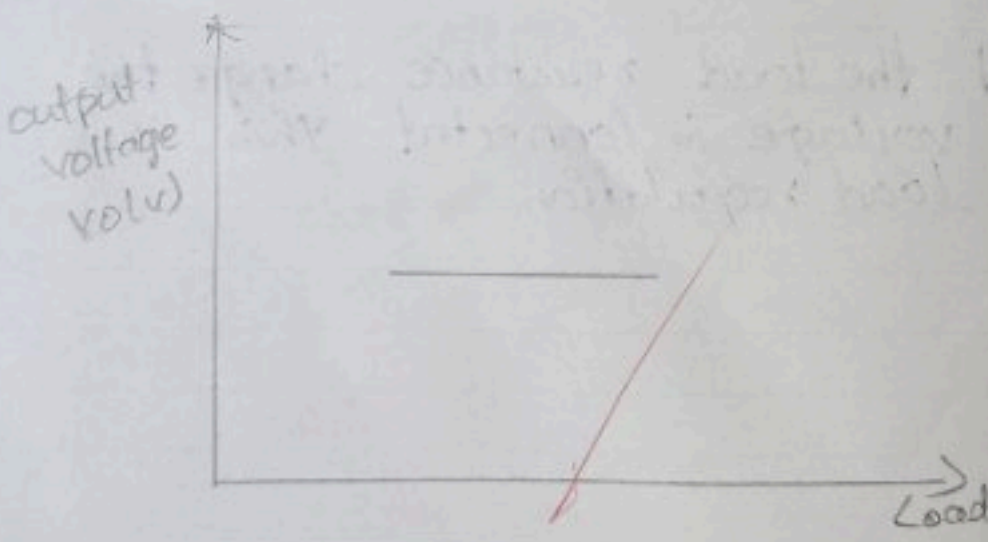
Load Regulation



Tabulation

S.NO	Load Resistance $R_L(\Omega)$	Output voltage $V_o(V)$
1	100Ω	+5.19 V
2	200Ω	+5.20
3	300Ω	+5.20
4	400Ω	+5.20
5	500Ω	+5.20
6	600Ω	+5.21

Model diagram



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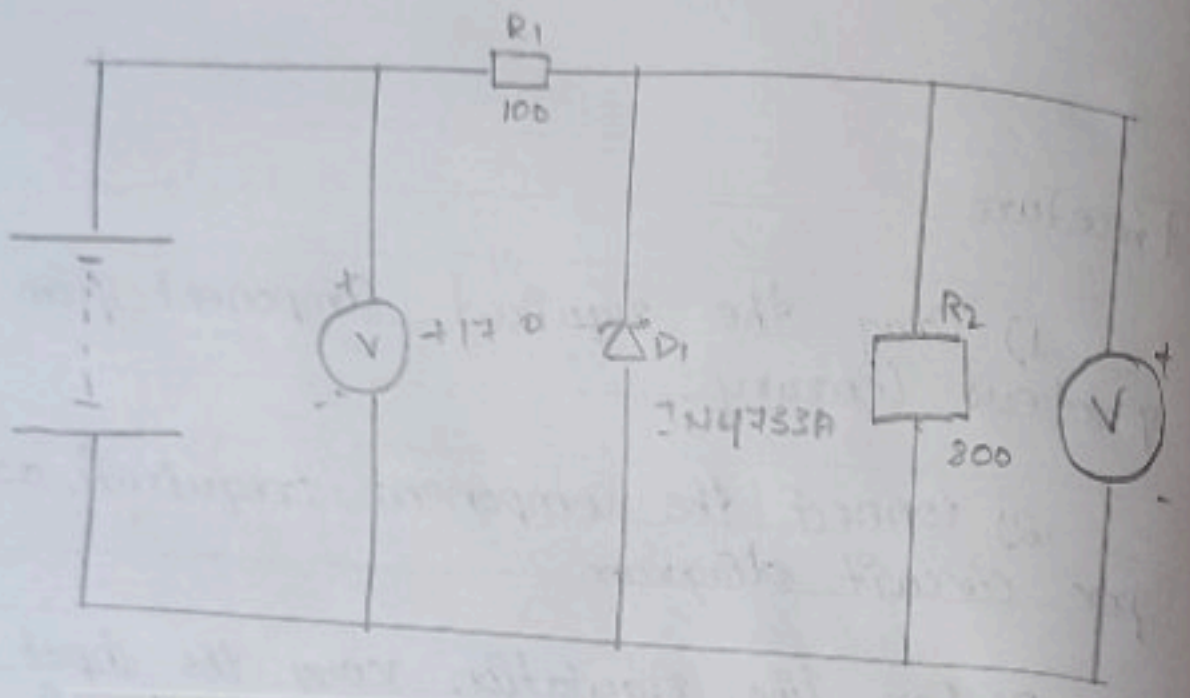
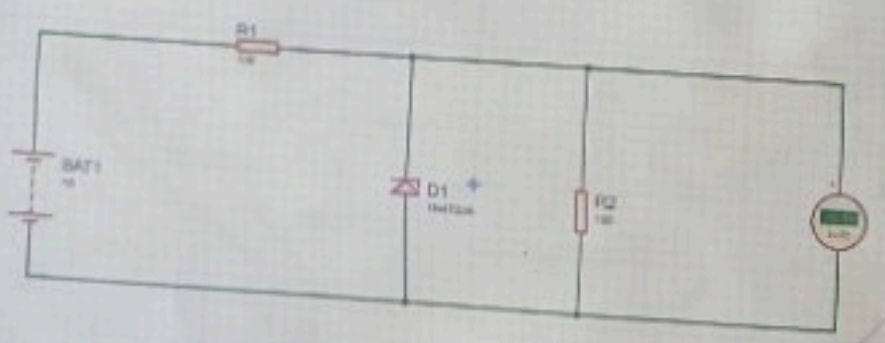
Result:

The stimulation of line load regulation operation of zener diode successfully completed.

output-